Questionnaire for the EOSC-Pillar "National Initiatives" Survey

Questionnaire and Dataset Description

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Related documents:

- **Dataset**: Bodlos, A., Hönegger, L., Kaczmirek, L., Beckmann, V., Breton, V., Romier, G., van Wezel, J., Streit, A., Stevanovic, U., Galeazzi, F., Tanlongo, F., van Nieuwerburgh, I. (2020). EOSC-Pillar "National Initiatives" Survey. Vienna: AUSSDA. doi: 10.11587/VOSVGK.
- **List of targets**: Bodlos, A., Hönegger, L., Kaczmirek, L., Beckmann, V., Breton, V., Romier, G., van Wezel, J., Streit, A., Stevanovic, U., Galeazzi, F., Tanlongo, F., van Nieuwerburgh, I. (2019). List of targets for the EOSC-Pillar "National Initiatives" Survey: E-infrastructures, research infrastructures, funding bodies and universities in Austria, Belgium, France, Germany and Italy. Vienna: AUSSDA. doi: 10.11587/VOSVGK.
- **Frequency Analysis**: Bodlos, A., Hönegger, L., Kaczmirek, L., Beckmann, V., Breton, V., Romier, G., van Wezel, J., Streit, A., Stevanovic, U., Galeazzi, F., Tanlongo, F., van Nieuwerburgh, I. (2020). Results of the EOSC-Pillar "National Initiatives" Survey: Frequency Analysis. Vienna: AUSSDA. doi: 10.11587/VOSVGK.
- **Summary Report**: Bodlos, A., Hönegger, L., Kaczmirek, L., Beckmann, V., Breton, V., Romier, G., van Wezel, J., Streit, A., Stevanovic, U., Galeazzi, F., Tanlongo, F., van Nieuwerburgh, I. (2020). Summary report of the EOSC-Pillar "National Initiatives" Survey. Zenodo. doi: 10.5281/zenodo.3937318.

Version control:

Version 2.0 contains several minor corrections of version 1.0 of the questionnaire (including corrections of value labels). Besides, version 2.0 contains major updates on additional information on the datasets and data curation processes.

Acknowledgements:

We are grateful to the input and feedback that stakeholders, colleagues and testers gave during two rounds of feedback, cognitive pretesting and other stages of the research design.

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Introduction

This document contains the questionnaire of the "National Initiatives" Survey that was conducted as part of the EOSC-Pillar H2020 project in fall 2019 by Anita Bodlos, Lisa Hönegger, Lars Kaczmirek, Volker Beckmann, Vincent Breton, Geneviève Romier, Jos van Wezel, Achim Streit, Uros Stevanovic, Fulvio Galeazzi, Federica Tanlongo, and Inge Van Nieuwerburgh. The survey is a cross-section study in the five countries Austria, Belgium, France, Germany, and Italy. In addition, this document contains a brief overview of the survey design and a detailed description of the datasets.

For a discussion on how to interpret the data, the scope and limitations of this study and many other methodological aspects, please also see the related documents "Frequency analysis"[1], "Summary report"[2] and "List of targets"[3] (see p. 2).

[1] Bodlos, A., Hönegger, L., Kaczmirek, L., Beckmann, V., Breton, V., Romier, G., van Wezel, J., Streit, A., Stevanovic, U., Galeazzi, F., Tanlongo, F., van Nieuwerburgh, I. (2020). Results of the EOSC-Pillar "National Initiatives" Survey: Frequency Analysis. Vienna: AUSSDA. doi: 10.11587/VOSVGK.

[2] Bodlos, A., Hönegger, L., Kaczmirek, L., Beckmann, V., Breton, V., Romier, G., van Wezel, J., Streit, A., Stevanovic, U., Galeazzi, F., Tanlongo, F., van Nieuwerburgh, I. (2020). Summary report of the EOSC-Pillar "National Initiatives" Survey. Zenodo. doi: 10.5281/zenodo.3937318.

[3] Bodlos, A., Hönegger, L., Kaczmirek, L., Beckmann, V., Breton, V., Romier, G., van Wezel, J., Streit, A., Stevanovic, U., Galeazzi, F., Tanlongo, F., van Nieuwerburgh, I. (2019). List of targets for the EOSC-Pillar "National Initiatives" Survey: E-infrastructures, research infrastructures, funding bodies and universities in Austria, Belgium, France, Germany and Italy. Vienna: AUSSDA. doi: 10.11587/VOSVGK.

Survey Design

This questionnaire is the basis of a cross-section study aiming at landscaping national initiatives of open research data and services with relevance to the European Open Science Cloud (EOSC) in five European countries (Austria, Belgium, France, Germany and Italy). The target population consists of four target groups that are crucial to the implementation of EOSC:

- e-infrastructures (including data repositories, data archives, data/computing centres, grid infrastructure, HPC's, NREN's and national museums with digital archives),
- research infrastructures of national relevance,
- universities and
- funding bodies.

The aim of the survey was to obtain representative results for each target group. The survey was developed and conducted following best practices in online survey methodology. For each country and target group, we compiled a comprehensive list of organisations and individual contact details using public databases and listings (e.g. records provided by government institutions on universities and research infrastructures of national relevance, records of national ESFRI nodes as well as records of repositories and HPCs).

For every organisation, we identified the contact details of one expert (most likely a scientific or managing director, depending on the type of organisation). These experts received an invitation to participate in the survey as representatives of their organisation and were asked to provide answers and to gather relevant information within the organisation if necessary (for more information, please see the related document "List of Targets"[4]).

Structure of the Questionnaire

All respondents received a common core ("Core Module") of questions addressing their perceptions of EOSC along with questions about the participating organisation. Additionally, we tailored sets of questions to each target group. All respondents received only the questions relevant to them according to our classification of organisations in one of the target groups.

The questionnaires for funding bodies, universities and research infrastructures are comparably short (less than 25 questions each). These institutions mainly received questions concerning policies of their institutions that are relevant to open science and EOSC.

[4] Bodlos, A., Hönegger, L., Kaczmirek, L., Beckmann, V., Breton, V., Romier, G., van Wezel, J., Streit, A., Stevanovic, U., Galeazzi, F., Tanlongo, F., van Nieuwerburgh, I. (2019). List of targets for the EOSC-Pillar "National Initiatives" Survey: E-infrastructures, research infrastructures, funding bodies and universities in Austria, Belgium, France, Germany and Italy. Vienna: AUSSDA. doi: 10.11587/VOSVGK.

E-infrastructures are the target group in our definition that provide services relevant to the EOSC catalogue and whose maturity level we aim to assess. Therefore, e-infrastructures received the most detailed questionnaire. To allow for a more fine-grained analysis, we asked the respondents of e-infrastructures which services they offer to the research community. For defining the services of e-infrastructures, we built upon previous work by the European Commission and EOSC pilot:

> *"The Cloud can be understood as the combination of three interdependent elements: the data infrastructures which store and manage data; the high-bandwidth networks which transport data; and the ever more powerful computers which can be used to process the data."*[5]

When compiling the experts for our targeted organisations, we aimed at identifying respondents that are experts in charge of one of the services according to this definition. If an organisation offered multiple services, we aimed at addressing one expert per service (e.g. the heads of the departments rather than the head of the organisation). When this was not possible and one respondent is an expert for multiple services, he or she received the questions multiple times (e.g. respondents are asked all questions once for "data infrastructures which store and manage data" and once for the "powerful computers which can be used to process the data").

The survey was implemented with the open source software LimeSurvey (https://www.limesurvey.org). The programming of the questionnaire was outsourced to a company specializing on LimeSurvey consulting. Information for programmers or users of the dataset describing the questionnaire logic, filter or recodings are marked with "*" or in "[]" below and were not shown to respondents. Similarly, the question number (e.g. "F1U1R1E8.") and the answer code (e.g. „1 -„) was not shown to respondents. The term „block" indicates a set of questions that are presented on the same webpage. The information to which of the four target groups a respondent belongs was stored and preloaded in the variable "attribute_2" in LimeSurvey as part of the contact details.

[5] European Commission (2016): European Cloud Initiative - Building a competitive data and knowledge economy in Europe, available at: https://ec.europa.eu/transparency/regdoc/rep/1/2016/EN/1-2016-178-EN-2-1.PDF, last accessed: 6 November 2019, p. 2.
see also: Terrovitis, Manolis; Tsiavos Prodromos; Oster, Per; Sandberg, Marie; Gheller, Claudio; Pineda, Oriol; Segers, Philippe (15 May 2015): D2.1 Draft Stakeholder Map (deliverable D2.1 by EOSC pilot), available at: https://eoscpilot.eu/sites/default/files/eoscpilot-d2.1.pdf, last accessed: 6 November 2019, p. 9.

Pseudonymization of the datasets

In order to minimize the risk of identification of respondents, we have taken several precautions and modifications to the dataset in line with best practice examples of how to comply to the GDPR. In more detail, we have taken the following steps:

1) We dropped all information that unambiguously identifies respondents or their organisation (names of respondents and their organisation, email addresses) including the questions R301/R311/R322. "Please provide a link to a website that describes this service. If you do not have a link, what is the name of the service?".

2) We dropped all open-ended answers before publication as they may contain information that unambiguously identifies respondents. In more detail, we deleted the following variables:

- all information gathered from the text field of "other, please specify" categories
- all comments that respondents could give at the end of a module
- individual questions including open text fields:
    - F18U70R70. "In what way is your organisation supporting EOSC?"
    - F10U10R17E17. "What do you expect from EOSC?"
    - F14. "Does your organisation maintain a roadmap of the infrastructures you fund? How many years does this roadmap cover?". We dropped the open text fields containing the number of years of roadmaps
    - E95. "In what year was your organisation founded initially?"
    - E85. "What is the name of this organisation you are part of or related to?"
    - E83. "Please indicate one or more of your services that you would like to become part of EOSC."
    - E82. "Which services does your organisation aim at introducing within the next five years? What are the next steps of innovation of your institution?"
    - E23_1/2/3. "Please describe which policies, procedural and/ or technical barriers limit the expansion of your services to further user groups:"
    - E81_1/2/3. "From which additional services would the users of your organisation benefit?"
    - E661 "Please list the most important standardized/ controlled vocabularies your organisation uses in your metadata."
    - E76. "How many datasets are accessible online?"

- E77. "Please think about datasets that are reused frequently. What characteristics are important for a frequent reuse of datasets?"
- E2. "What was your organisation's total budget (in €) in 2018?" and the follow-up question E3 on the individual services' budget
- E4. "How many employees did your organisation have in 2018? Please provide a number in full time equivalents." and the follow-up question E5 on the individual services.
- E6. "What was the estimated gender distribution in your staff in 2018? Please provide the percentage of women." and the follow-up question E7 on the individual services.

3) In addition, we dropped closed-ended questions that contain information that would facilitate the identification of respondents. On an aggregate level, this information is available in the related document "Frequency analysis"[6].

- We deleted the variables holding the information on which roadmap an e-infrastructure (E24) or research infrastructure (R18) participates. We recoded this information to a variable indicating whether the organisation is part of any roadmap.
- E37. "Does your organisation participate in the following European organisations?" We recoded the response categories so that the variable E37 now captures whether respondents participate in any organisation (including "other"), or not ("none") or whether they ticked "77 - don't know".

4) Further measures to minimize the risk of identification of respondents:

- We deleted the information to which countries the funding bodies belong.
- We publish the more sensitive questions capturing perceptions in a separate dataset. Thereby, we ensure that even if an organisation can be identified, the respondent's perceptions cannot be connected to this information. The name of the dataset capturing perceptions is 100083_da03_en_v1_0.

[6] Bodlos, A., Hönegger, L., Kaczmirek, L., Beckmann, V., Breton, V., Romier, G., van Wezel, J., Streit, A., Stevanovic, U., Galeazzi, F., Tanlongo, F., van Nieuwerburgh, I. (2020). Results of the EOSC-Pillar "National Initiatives" Survey: Frequency Analysis. Vienna: AUSSDA. doi: 10.11587/VOSVGK.

Data curation

Missing values:

Respondents were set to "missing" when they did not answer any question of a module or when they never received a question (e.g. due to filter variables). If respondents chose to skip individual questions, they were coded as "9999 – no answer".

"Other" category:

Respondents could only choose the response category "other", if they typed text in the accompanying text field. For the dataset, these answers were recoded to a numeric variable indicating whether respondents did fill some text in this field and tick the "other"-option ("yes"), or not ("no"). In addition, we automatically made some minor recoding of the strings in the text field: If respondents typed "no answer" or "not applicable", we recoded these responses using the respective values (99 – "not applicable" and 9999 – "no answer"). As a consequence, variables capturing the "other"-options sometimes comprise the value "not applicable" although respondents were not given this option.

Exclusive categories:

Exclusive categories of multiple choice questions usually were recoded either into each variable capturing the response categories or into variables that mimic filter variables:

An example for the recoding of an exclusive response category into all other categories is question F12. In this question, respondents could choose the response category "not applicable". However, if respondents chose this answer, they could not choose any other response category, hence, this category is an "exclusive category". Therefore, all other variables capturing the response categories of variable F12 contain the value "99 – n.a.". This information is provided next to the respective response category in the questionnaire.

An example for a recoding of exclusive categories into the style of filter variables is question F14. In this question, we asked representatives of funding bodies (among other things) "Does your organisation maintain a roadmap of the infrastructures you fund?" Respondents could choose the response categories "No", "Yes, aligned to a European roadmap. "Yes, aligned to a national roadmap" and "Yes, we maintain a roadmap according to our own specifications." The response category "no" is exclusive (as respondents cannot have no roadmap and a roadmap). In this case, we recoded the information of this variable to another variable indicating whether the respondents maintain any roadmap (hence, whether they ticked any of the "yes" responses) or not. If respondents did indicate to maintain any roadmap, the following variables indicate

which roadmaps respondents maintain. If we recoded variables in this way, we give the information in the programmer information.

Variable names:

The names of most variables in the dataset are identical to or can be derived from the question IDs as listed in this document. An exception are the questions of the core module (CM) that either have speaking names ("EOSCfamiliar" and "FAIRfamiliar") or consecutive numbers (CM_1, CM2…). If the variable name deviates from the question ID and the pattern used for naming variables (described below), this information is provided in the programming information for the respective question.

For multiple choice questions, the individual response categories usually form separate variables. For funding bodies, universities and research infrastructures, these variables have names in the style of *questionID_responseCategory*. E.g. variable F12_1 contains the first response category of question F12. The names for variables dedicated to e-infrastructures follow a different pattern as these questions may have been asked several times (depending on the number of services e-infrastructures provide). How these variables for the responses of e-infrastructures are named is described below along with the dataset structures.

Values and value labels:

Throughout the dataset, the following values and value labels were used frequently (but not always):

Value	Value label	Comment
1	no	Frequently used for yes/no and multiple choice questions
2	yes	Frequently used for yes/no and multiple choice questions
77	don't know	
88	other	
99	not applicable/ n.a.	
9999	no answer	respondent skipped the question
missing		Respondent did not receive this question (because s/he dropped out or due to filter questions)

Table 1 Frequently used value labels

Dataset structure

The information gathered by this questionnaire is published in three datasets. In short, the datasets contain the following information:

- Dataset 100083_da01_en_v1_0 contains all non-perceptive information in wide format (organisation level for e-infrastructures).
- Dataset 100083_da02_en_v1_0 contains a subsample of questions dedicated to e-infrastructures of the dataset 100083_da01_en_v1_0 reshaped to a long format (service level, stacked dataset).
- Dataset 100083_da03_en_v1_0 contains all perceptive information in wide format.

The content for each dataset is described in detail below:

Dataset 100083_da01_en_v1_0:

This dataset contains the information of all variables in the questionnaire except for the variables capturing perceptions (which are published separately in the dataset 100083_da03_en_v1_0, see the section on pseudonymization).

The format of this dataset is wide, or on the "organisation-level" as each observation refers to one organisation. Therefore, questions for e-infrastructures that refer to a specific service each form a separate variable. For instance, the question E25. "Does your organisation charge users/clients for services?" was asked once to respondents who indicated that they "store and manage research data" (SMD), once to respondents who offer high-bandwidth networks (HBN) and once to high-performance computing centres (HPC). As a consequence, there are three variables for question E25 in the dataset, one per service. To which service a variable refers is captured in the variable label ("SMD", "HBN" or "HPC"). In addition, this information is captured in the digit following the first underscore after the question ID ("1" for SMD, "2" for HBN, "3" for HPC). For instance, the variable E25_1 refers to SMDs, variable E25_2 to HBNs and E25_3 to HPCs. All single choice variables follow this pattern.

Multiple choice questions are then followed by another underscore and a third digit indicating the response category. An example may illustrate this logic:

- E19_1_1 was asked to SMDs and refers to the first response category of question E19.
- E19_1_2 was asked to SMDs and refers to the second response category of question E19.
- E19_2_1 was asked to HBNs and refers to the first response category of question E19.

- E19_2_2 was asked to HBNs and refers to the second response category of question E19.
- E19_3_1 was asked to HPCs and refers to the first response category of question E19.
- E19_3_2 was asked to HPCs and refers to the second response category of question E19.

Hence, in this dataset, the variable names dedicated to questions of e-infrastructures' services follow the pattern *questionID_Service* or *questionID_Service_responseCategory*.

Several questions were asked to e-infrastructures only once either because they are valid for the entire organisation and not only for one service or because they were asked only to repositories (SMDs, and not to HPCs or HBNs). To distinguish these variables from the variables asked several times, these variables do not contain any underscores. We provide the names of these variables in the programming information at the respective questions in this document. Thus, in contrast to the variables dedicated to funding bodies, universities and research infrastructures, the underscores in variables for e-infrastructures always refer to services (and frequently also to response categories of multiple choice questions), but never only to response categories of multiple choice questions. The variable "ID_org" is a genuine ID-variable for this dataset.

Dataset 100083_da02_en_v1_0:

This dataset is a reshaped subset of the dataset 100083_da01_en_v1_0 and refers to the "service level" (and not to the "organisation level"). This dataset only contains non-perceptive information for questions of e-infrastructures that refer to the service level and hence were asked to respondents once per service they provide. The main difference to the dataset 100083_da01_en_v1_0 is that information for individual services is not captured in separate variables, but in one single variable. Thus, the format of the dataset is long or on a "service level". We also used the term "stacked dataset" for this dataset. We used this dataset for several analyses in the "Frequency Analysis" and in the "Summary report". In a nutshell, the information provided in the dataset 100083_da01_en_v1_0 and 100083_da02_en_v1_0 is identical but presented in a different format to enhance reusability.

The variable names follow the same pattern as described above for the dataset 100083_da01_en_v1_0. However, the variables usually referring to SMDs contain the information of all services combined. So, the variable E19_1_1 does not only refer to SMDs, but also contains information on HBNs and HPCs. Consequently, the variables E19_2_1 and E19_3_1 do not exist. To which service an observation relates is captured in the variable "service". For multiple choice questions, the digit after the second underscore again identifies

the response category. The variable "ID_org" in this dataset is identical to the ID variable in the dataset 100083_da01_en_v1_0. However, as the dataset was reshaped to a long format, this ID variable does not uniquely identify observations. Therefore, we created another ID variable ("ID_service") that indeed is a genuine ID variable for this dataset. All variables that do not refer to services provided by e-infrastructures are excluded from this dataset.

Dataset 100083_da03_en_v1_0:

This dataset contains all information gained by questions asking the perception of respondents. All questions of the Core Module are part of this dataset as well as several other questions (E54, E63, E69). The dataset structure is identical to the structure of the dataset 100083_da01_en_v1_0, hence, in a wide format. Thus, each observation refers to one organisation (and not a service). The ID variable in this dataset is called "ID_perc" and was randomly assigned. Hence, the dataset 100083_da03_en_v1_0 cannot be connected to any of the other datasets.

Variables in addition to the questions of the questionnaire:

In addition to the variables described in the questionnaire, the following variables are part of one or more datasets

Variable name	Variable label	Value label	Description	Part of which dataset
doi	Digital Object Identifier	*None (string)*	DOI of the dataset	all
version	AUSSDA archive version	*None (string)*	Version of the dataset	all
format	Format	*None (string)*	Format of the dataset (wide or long)	all
target_group	Target group	1 "FB" 2 "Uni" 3 "RI" 4 "E-I"	Target group to which the observation refers; FB – funding bodies, Uni – universities, RI – research infrastructures, E-I e-infrastructures	all
country	Country	1 "AT" 2 "BE" 3 "DE" 4 "FR" 5 "IT"	Country where the organisation is located; AT – Austria, BE – Belgium, DE – Germany, FR – France, IT - Italy	all
service	Service provided by the e-infrastructure	1 "SMD" 2 "HBN" 3 "HPC"	Service to which the observation refers	100083_da02_en_v1_0
ID_org	ID: organisation level	*None*	ID variable for the non-perceptive dataset (cannot be connected to the variable ID_per), wide format/organisation level	100083_da01_en_v1_0, 100083_da02_en_v1_0
ID_service	ID: service level	*None*	ID variable for the non-perceptive dataset (cannot be connected to the variable ID_per) in the long-format (service level	100083_da02_en_v1_0
ID_perception	ID: perception dataset	*None*	ID variable for the dataset containing perceptive information (cannot be connected to the variables ID_org or ID_service)	100083_da03_en_v1_0

Table 2 Variables not described in the questionnaire

Ethical and legal aspects

Survey participants received a detailed information sheet on the processing and re-use of the provided data including the aim of the survey and aspects on data collection, data processing, the responsible entities and the participants' rights regarding their personal data. This information was part of the page preceding the survey. Accessing the survey was only possible after respondents agreed to the terms as specified in the information sheet/ consent form. The information sheet/consent form used for the survey is as follows.

Information for participants in the EOSC-Pillar study with regard to the processing and re-use of data.

Title of Study: "National Initiatives" Survey

This project has received funding from the European Union's Horizon 2020 Research and Innovation Programme under grant agreement No. 857650.

You were selected to participate in the online survey on national and thematic initiatives on open research data and services with relevance to the European Open Science Cloud (EOSC). EOSC has been proposed in 2016 by the European Commission as part of the European Cloud Initiative to build a competitive data and knowledge economy in Europe. The European Open Science Cloud (EOSC) was officially launched in November 2018 at the University of Vienna.

This survey is being conducted by the University of Vienna, the Karlsruhe Institute of Technology (KIT) and the French National Centre for Scientific Research (CNRS). The survey will be conducted in Austria, Germany, France, Italy and Belgium.

The aim of this survey and at the same time the purpose of the data processing is to identify and describe national and thematic initiatives on open research data and services.

Your responses may help us learn more about the state of the art of national and thematic initiatives with regard to open science and open data. The results of the study will be published in a report which will be openly accessible. This knowledge will form the basis of strategic orientation and future policy developments to offer Europe's 1.7 million researchers and 70 million science and technology professionals a virtual environment to access, process, store, share, transport and re-use research data – ultimately to foster progress and innovation in the European research landscape, benefitting science and society.

Your participation in this survey is voluntary. You may refuse to take part in the survey or exit the survey at any time without penalty. You are free to decline to answer any particular question you do not wish to answer for any reason.

Are there any risks when taking part in this study?

We do not anticipate any risks for you from participating in this study.

Who is the responsible party for processing data?

The party responsible for processing data is:
University of Vienna
Universitätsring 1
1010 Vienna
E-mail: dsba@univie.ac.at

Which data do we collect and process?

We collect and process your contact details (name, e-mail address, position in your institution, name of your institution) and your answers to the questions of the survey.

How do we collect and process your data?

Provided contact details and your answers will be kept separate from your answers and can only be connected via an additional key to ensure data protection and security. Data collection and processing is carried out in accordance with the GDPR, in particular in accordance with Art. 6 para 1 lit. f. GDPR.

The servers hosting the survey are located in Germany. Once the survey is concluded, the data will be processed by authorized employees of the University of Vienna (AUSSDA), the Karlsruhe Institute of Technology (KIT) and the French National Centre for Scientific Research (CNRS). Persons who have access to your data are obligated to non-disclosure.

Your answers will be used in aggregated form in publications. In addition, and if you agree, your information can be quoted in research outputs without naming you as the source.
AUSSDA will curate and pseudonymize the data, make it comply to the FAIR principles and archive and preserve it.

The data will be processed, stored, and distributed to GARR (Italy) for back up purposes.

In pseudonymized form, the responses to the survey will be available via the AUSSDA data repository with specific access conditions. These conditions are determined on the basis of the data and level of curation/pseudonymization. The data will then be published as Open Access or Scientific Use file, in compliance with GDPR standards.

Several EOSC related projects are cooperating to reduce the required amount of repeated data collection activities and to harmonize surveys across countries. Your responses may be linked to answers from your organization in other surveys in the context of open research data and services. This will reduce your respondent burden.

What rights do you have regarding your data?

You have the right to request information about your stored data, its origin, its recipients, and the purpose of its collection at no charge. You also have the right to request that it be corrected, blocked, or deleted. You can contact the responsible party at

any time using the address given at the end of this statement, if you have further questions about the issue of privacy and data protection.

Pursuant to Article 77 GDPR, you have the right to complain to a supervisory authority.

In Austria, the supervisory authority is the Data Protection Authority, Barichgasse 40-42, 1030 Vienna, telephone: +43 1 52 152-0, e-mail: dsb@dsb.gv.at, website: dsb.gv.at

In Italy: Garante per la protezione dei dati personali - Ufficio relazioni con il pubblico, piazza Venezia n. 11, Roma, Telefono: (+39) 06.69677.2917, E-mail: urp@gpdp.it, website: https://www.garanteprivacy.it/home/footer/contatti#urp

In France: Commission Nationale de l'Informatique et des Libertés 3 Place de Fontenoy, TSA 80715, 75334 PARIS CEDEX 07 Téléphone : +33 (0)1 53 73 22 22, website: https://www.cnil.fr/fr

In Germany: Der Landesbeauftragte für den Datenschutz und die Informationsfreiheit, Königstraße 10 a, 70025 Stuttgart, Telefon +49 (0)711 61 55 41 0, E-mail: poststelle@lfdi.bwl.de, website: https://www.baden-wuerttemberg.datenschutz.de, oder einer der weiteren Landesbeauftragten für den Datenschutz im Bund nach Wahl

In Belgium: De Gegevensbeschermingsautoriteit / L'Autorité de protection des données, webiste: https://www.gegevensbeschermingsautoriteit.be/verzoek-klacht-indienen, GBA Drukpersstraat 35, 1000 Brussel, tel +32 (0)2 274 48 00, e-mail: contact@apd-gba.be

ELECTRONIC CONSENT: Please select your choice below. You may print a copy of this consent form for your records. Clicking on the “Agree” button indicates that

- You have read the above information
- You voluntarily agree to participate

□ I confirm that I have read the information with regard to the processing and re-use of data

****Questions in LimeSurvey following the consent form:****

L2 I agree that my verbatim responses can be quoted in reports and research outputs. These quotes will not contain your or your organisation's name to protect your privacy.

□ Yes

□ No

L1 I agree that I may be contacted for follow-up questions for clarification and elaboration purposes

□ Yes

□ No

QUESTIONNAIRE

The following pages contain the questionnaire of the EOSC-Pillar "National Initiatives" Survey.

Questionnaire for all respondents (Core Module)

*** Programmer information: respondents should only receive these questions if L1 == "Yes". ***

*** Block G1: personal information ***

[Title of the block:] Personal information

[Description of the block:] We aim at assessing the landscape of services relevant for EOSC as thoroughly as possible. Therefore, your expertise is very important. Please provide your name and e-mail address to allow us to contact you if we have follow-up questions.

F1U1R1E8. What is your first name?

[short free text]

F2U2R2E9. What is your last name?

[short free text]

F3U3R3E10. What is your e-mail address?

[short free text]

F4U4R4E11. Which term describes your position in your organisation best?

[long free text]

*** Block B_G2: EOSC ***

[Title of the block:] The European Open Science Cloud (EOSC)

[Description of the block:] In the following questions, we would like to ask you about your perception of the European Open Science Cloud (EOSC) and how it affects your organisation.

F5U5R12E12. Considering everything you know about EOSC, how familiar are you with EOSC?

["Hilfetext":] Here is more information on EOSC: https://www.eosc-portal.eu/about/faqs

[single choice, variable name: EOSCfamiliar]

- ◻ 1 - very familiar
- ◻ 2 - familiar
- ◻ 3 - not very familiar
- ◻ 4 - not familiar at all

F6U6R13E13. How does or will EOSC affect your organisation and/ or your strategic plans?

[single choice, Condition: ask only if "EOSCfamiliar" is NOT "4 - not familiar at all", variable name: CM_2]

- ◻ 1 - very much
- ◻ 2 - somewhat
- ◻ 3 - not very much
- ◻ 4 - not at all
- ◻ 77 - don't know

F7U7R14E14. Will your organisation benefit from the implementation of EOSC?

[single choice, Condition: ask only if "EOSCfamiliar" is NOT "4 - not familiar at all", variable name: CM_3]

- ◻ 1 - very much
- ◻ 2 - somewhat
- ◻ 3 - not very much
- ◻ 4 - not at all
- ◻ 77 - don't know

U94R94E94. Is your organisation already contributing to EOSC?

[single choice, Condition: ask only if "EOSCfamiliar" is NOT "4 - not familiar at all" AND attribute_2 == ("research_infrastructure" OR "e-infrastructure" OR "university"), variable name: CM_4]

- ◻ 2 - yes
- ◻ 1 - no
- ◻ 77 - don't know

U8R15E15. How likely is it that your organisation will contribute to EOSC?

[single choice, Condition: ask only if "EOSCfamiliar" is NOT "4 - not familiar at all" AND attribute_2 == ("research_infrastructure" OR "e-infrastructure" OR "university") AND U94R94E94 is NOT "2 - yes"), variable name: CM_5]

- ◻ 1 - very likely
- ◻ 2 - likely
- ◻ 3 - not very likely
- ◻ 4 - not likely at all
- ◻ 77 - don't know

F18U70R70. In what way is your organisation supporting EOSC?

[long free text, Condition: ask only if "EOSCfamiliar" is NOT "4 - not familiar at all" AND attribute_2 == ("research_infrastructure" OR "university" OR "funding_body" AND U94R94E94 == "yes"]

F9U9R16E16. How beneficial is/ will be contributing to EOSC for your organisation?

[single choice, Condition: ask only if "EOSCfamiliar" is NOT "4 - not familiar at all", variable name: CM_7]

- ◻ 1 - very beneficial
- ◻ 2 - somewhat beneficial
- ◻ 3 - not very beneficial
- ◻ 4 - not beneficial at all
- ◻ 77 - don't know

F10U10R17E17. What do you expect from EOSC?

[long free text; Condition: ask only if "EOSCfamiliar" is NOT " 4 - not familiar at all"]

F17U12R20E62. How familiar are you with the FAIR principles regarding data? FAIR means Findable, Accessible, Interoperable and Reusable.

[all text above is part of the question (not of the "Hilfetext"), single choice, variable name: FAIRfamiliar]

- ◻ 1 - very familiar
- ◻ 2 - familiar
- ◻ 3 - not very familiar
- ◻ 4 - not familiar at all

CG2. Would you like to give a comment on any of the questions above?

[long free text]

Questionnaire for funding bodies

*** Programmer Information: Only respondents for which attribute_2 == "funding_body" will receive the question of the section "Questionnaire for Funding bodies" ***

*** Block B_F1: business information ***

[Title of the block:] Grant regulations

[Description of the block:] In the next questions, we would like to ask you about the kind of grants you offer, their regulations and recipients.

F11. What does your organisation fund?

[multiple choice]

- ◻ 1 - human resources
- ◻ 2 - hardware
- ◻ 3 - software
- ◻ 4 - capital expenditure (capex) at large
- ◻ 5 - operational expenditure (opex) at large
- ◻ 6 - project based resources
- ◻ 88 - other, please specify:

F12. Does your organisation have rules for granting funds for e-infrastructures or research infrastructures based on the following aspects?

[multiple choice]

- ◻ 1 - discipline of the infrastructure's users
- ◻ 2 - geographical location of the infrastructure's users
- ◻ 3 - selection by a competitive process
- ◻ 4 - affiliation of the infrastructure's users
- ◻ 88 - other, please specify:
- ◻ 99 - not applicable [exclusive category, recoded in all other categories]

F13. Do you require infrastructures you fund to provide the cost information about the services they offer?

[single choice]

- ◻ 1 - yes, always
- ◻ 2 - yes, for some grants
- ◻ 3 - no, but (maybe) in the future
- ◻ 4 - no
- ◻ 99 - not applicable

F14. Does your organisation maintain a roadmap of the infrastructures you fund? How many years does this roadmap cover?

Please tick the boxes and insert the number of years in the corresponding text field.

[all the text above is part of the question (not of the "Hilfetext"), multiple choice with comment (ideally the text fields has a length of ten or less characters), question was recoded: variable F14 captures whether respondents chose any affirmative answer. If respondents did so, the variables F14_2-F14_4 contain the information which option(s) they chose.]

- ◻ 1 – No [exclusive category]
- ◻ 2 - Yes, aligned to a European roadmap. This roadmap covers the following number of years:
- ◻ 3 - Yes, aligned to a national roadmap. This roadmap covers the following number of years:
- ◻ 4 - Yes, we maintain a roadmap according to our own specifications. This roadmap covers the following number of years:

F15. Does your organisation allow infrastructures who receive grants to spend funding on user support?

[multiple choice]

- ◻ 1 - We explicitly offer funding for user support.
- ◻ 2 - Grant regulations allow for spending funds on user support.
- ◻ 3 - Grant regulations prohibit spending funds on user support.
- ◻ 88 - other, please specify:
- ◻ 99 - not applicable [exclusive category, recoded in all other categories]

CF1. Would you like to give a comment on any of the questions above?

[long free text]

Block B_F2: Policies

[Title of the block:] Rules for funding

[Description of the block:] We would like to ask you about rules for funding you impose on grant receivers that concern EOSC and open science in general.

F16. Does your organisation impose rules for funding on grant recipients concerning the following aspects?

[array question]

	1 - no regulation	2 - encouraged but optional for grants	3 - mandatory for some grants	4 - mandatory for all grants	5 - not applicable
A - open access publications	☐	☐	☐	☐	☐
B - data management plans	☐	☐	☐	☐	☐
C - open research data	☐	☐	☐	☐	☐
D - long-term availability of research data	☐	☐	☐	☐	☐
E - compliance of data to the FAIR principles	☐	☐	☐	☐	☐
F - publication of data in a repository	☐	☐	☐	☐	☐
G - publication of data in a certified repository	☐	☐	☐	☐	☐
H - other, please specify:	☐	☐	☐	☐	☐

CF2. Would you like to give a comment on the question above?

[long free text]

Questionnaire for universities

*** Programmer Information: Only respondents for which attribute_2 == "university" will receive the questions of the section "Questionnaire for universities" ***

[Title of the block:] EOSC and open science

[Description of the block:] In the next questions, we would like to ask you about expected benefits from EOSC and rules on open science at your organisation.

Block B_U1: Policies

U13 What research discipline(s) from your organisation will benefit mostly from EOSC?[7]

["Hilfetext":] Please follow this link for information on the classification of disciplines: https://en.wikipedia.org/wiki/Fields_of_Science_and_Technology

[multiple choice]

- ☐ 1 - Natural Sciences
- ☐ 2 - Engineering and Technology
- ☐ 3 - Medical and Health Sciences
- ☐ 4 - Agricultural Sciences
- ☐ 5 - Social Sciences
- ☐ 6 - Humanities
- ☐ 88 - Other, please specify:

[7] Source of categories (unchanged): Organisation for Economic Co-operation and Development (2007): Revised Field of Science and Technology Classification in the Frascati Manual; available at: https://www.oecd.org/science/inno/38235147.pdf; last accessed 6 November 2019.

U11. Some organisations have developed rules related to their work processes, whereas others have not. Has your organisation developed informal or formal regulations or publicly available policies that address the following aspects?

[array question]

	1 - no regulation	2 - informal regulation	3 - formal/ written regulation	4 - publicly available policy	99 - not applicable
A - open access publications	☐	☐	☐	☐	☐
B - research data management (RDM)	☐	☐	☐	☐	☐
C - open research data	☐	☐	☐	☐	☐
D - long-term availability of research data	☐	☐	☐	☐	☐
E - compliance of data to the FAIR principles	☐	☐	☐	☐	☐
F - publication of data in a repository	☐	☐	☐	☐	☐
G - publication of data in a certified repository	☐	☐	☐	☐	☐
H - other, please specify:	☐	☐	☐	☐	☐

CU1. Would you like to give a comment on any of the questions above?

[long free text]

Questionnaire for research infrastructures

*** Programmer Information: Only respondents with the attribute "research_infrastructure" will receive the questions of the section "Questionnaire for Research Infrastructures" ***

*** Block B_R1: General information ****

[Title of the block:] General information and policies

[Description of the block:] In the next section, we would like to ask you general questions about your organisation as well as rules and policies you may have developed.

R11. For which scientific discipline(s) does your organisation provide services?[8]

["Hilfetext":] Please follow this link for information on the classification of disciplines: https://en.wikipedia.org/wiki/Fields_of_Science_and_Technology

[multiple choice]

- ◻ 1 - Natural Sciences
- ◻ 2 - Engineering and Technology
- ◻ 3 - Medical and Health Sciences
- ◻ 4 - Agricultural Sciences
- ◻ 5 - Social Sciences
- ◻ 6 - Humanities
- ◻ 88 - Other:

R18. Is your organisation in an official roadmap?

[multiple choice; question was recoded: variable R18 captures whether respondents chose any affirmative answer.]

- ◻ 1 - no
- ◻ 2 - yes, national
- ◻ 3 - yes, ESFRI
- ◻ 4 - yes, other:
- ◻ 77 - don't know

[8] Source of the categories (unchanged): Organisation for Economic Co-operation and Development (2007): Revised Field of Science and Technology Classification in the Frascati Manual; available at: https://www.oecd.org/science/inno/38235147.pdf; last accessed 6 November 2019.

R19. Some organisations have developed regulations related to their work processes, whereas other organisations have not. Has your organisation developed informal or formal regulations or publicly available policies that address the following aspects?

[array question]

	1 - no regulation	2 - informal regulation	3 - formal/ written regulation	4 - publicly available policy	99 - not applicable
A - open access publications	☐	☐	☐	☐	☐
B - research data management (RDM)	☐	☐	☐	☐	☐
C - open research data	☐	☐	☐	☐	☐
D - long-term availability of research data	☐	☐	☐	☐	☐
E - compliance of data to the FAIR principles	☐	☐	☐	☐	☐
F - publication of data in a repository	☐	☐	☐	☐	☐
G - publication of data in a certified repository	☐	☐	☐	☐	☐
H - other, please specify:	☐	☐	☐	☐	☐

CR1. Would you like to give a comment on any of the questions above?

[long free text]

***** Block B_R2: services *****

[Title of the block:] Services

[Description of the block:] Some organisations offer services to the research community. Please indicate below whether your organisation offers these services.

R5 Does your organisation provide one or more of these services to the research community?[9]

[multiple choice]

- ◻ 1- knowledge-based resources such as collections, archives or research data
- ◻ 2- data and computing systems, and communication networks
- ◻ 3 - major scientific equipment or sets of instruments
- ◻ 88 - other research and innovation infrastructure of a unique nature essential to achieving excellence in research and innovation, please specify: [= "other"-option in Limesurvey]
- ◻ 4 - no, none of this list [exclusive category]

R30. Does your organisation offer data infrastructures which store and manage research data (e.g. archive and disseminate data)? [10]

[single choice]

- ◻ 2 - yes
- ◻ 1 - no

R301. Please provide a link to a website that describes this service. If you do not have a link, what is the name of the service?

[long free text, Condition: ask only if R30 == "2 - yes"]

[9] Categories based on the definition of "research infrastructures" used in: Official Journal of the European Union L 206/1, EN 8.8.2009, COUNCIL REGULATION (EC) No 723/2009, Article 2 (a); available at https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=OJ:L:2009:206:FULL&from=HR).

[10] Source of the questions R30, R31 and R32 (adapted): European Commission (2016): European Cloud Initiative - Building a competitive data and knowledge economy in Europe, available at: https://ec.europa.eu/transparency/regdoc/rep/1/2016/EN/1-2016-178-EN-2-1.PDF, last accessed: 6 November 2019, p. 2.

R31. Does your organisation offer high-bandwidth networks which transport research data?

[single choice]

- ◻ 2 - yes
- ◻ 1 - no

R311. Please provide a link to a website that describes this service. If you do not have a link, what is the name of the service?

[long free text, Condition: ask only if R31 == "2 - yes"]

R32. Does your organisation offer high-performance computing to process research data?

[single choice]

- ◻ 2 - yes
- ◻ 1 - no

R322. Please provide a link to a website that describes this service. If you do not have a link, what is the name of the service?

[long free text, Condition: ask only if R32 == "2 - yes"]

R21. Which term(s) fit(s) your organisation best?

[single choice]

- ◻ 1 - research infrastructure
- ◻ 2 - e-infrastructure
- ◻ 3 - both terms are equally valid
- ◻ 88 - neither term fits my organisation, the correct term is: [= "other"-option in Limesurvey"]

CR2. Would you like to give a comment on any of the questions above?

[long free text]

*** Questionnaire for e-infrastructures ***

*** Programmer Information: Only respondents for which attribute_2 == "e-infrastructure" will receive the questions of the section "Questionnaire for e-infrastructures" ***

Block B_E1: services

[Title of the block:] General information

[Description of the block:] In the next section, we would like to ask you general information on your organisation ({TOKEN:ATTRIBUTE_1}[11]), its services and its affiliation to other organisations.

E84. Is your organisation part of or related to another organisation which facilitates integrating your data and services into EOSC?

[single choice]

- ◻ 2 - yes
- ◻ 1 - no
- ◻ 77 - don't know

E85. What is the name of this organisation you are part of or related to?

[long free text, Condition: ask only if E84 == "2 - yes"]

E24. Is your organisation in an official roadmap?

[multiple choice; question was recoded: variable E24 captures whether respondents chose any affirmative answer]

- ◻ 1 - no
- ◻ 2 - yes, national
- ◻ 3 - yes, ESFRI
- ◻ 4 - yes, other, please specify:
- ◻ 77 - don't know

[11] The variable „attribute_1" in the address database contains the information on the respondents' organisation for which they are listed as experts.

E37. Does your organisation participate in the following European organisations?

[multiple choice, question was recoded: variable E37 captures whether respondents chose any affirmative answer (1 – EGI, 2 – EUDAT, 3 – PRACE, 88 – other), or not (99 – none) or whether they ticked "77 – don't know")]

- ☐ 1 - EGI
- ☐ 2 - EUDAT
- ☐ 3 - PRACE (Partnership for Advanced Computing in Europe)
- ☐ 88 - other, please specify:
- ☐ 99 - none
- ☐ 77 - don't know

E95. In what year was your organisation founded initially?

[short free text]

E1. Which service(s) does your organisation provide to the research community? [12]

[multiple choice]

- ☐ 1 - We offer data infrastructures which store and manage research data (e.g. archive and disseminate data). [variable name: E11]
- ☐ 2 - We offer high-bandwidth networks which transport research data. [variable name: E12]
- ☐ 3 - We offer high-performance computing which can be used to process research data. [variable name: E13]

E83. Please indicate one or more of your services that you would like to become part of EOSC.

[long free text]

E82. Which services does your organisation aim at introducing within the next five years? What are the next steps of innovation of your institution?

[long free text]

[12] Source of categories (adapted): European Commission (2016): European Cloud Initiative - Building a competitive data and knowledge economy in Europe, available at: https://ec.europa.eu/transparency/regdoc/rep/1/2016/EN/1-2016-178-EN-2-1.PDF, last accessed: 6 November 2019, p. 2.

CE1. Would you like to give a comment on any of the questions above?

[long free text]

***** Information for respondents *****

***

For respondents who indicated more than one answer in E1, we provide an instruction that they should think of one specific response category of E1 (i.e. service they provide) when answering the following blocks of questions. Hence, this information is shown two, maybe three times, depending on whether the respondents ticked two or three answers. (Please see the general programmer information for more explanations on the structure of the survey.)

Please provide this information as an own (empty) block of questions.

If the respondent only ticked one answer in E1: do not show any message.

If the respondent ticked more than one answer in E1, please take the first/second/third answer the respondent gave and write (in bold):
"**You indicated that you provide the following service: [*insert first/second/third ticked response category of E1*]. The following questions all refer to this service.**"

A smaller, less prominent version of this information is also included in the individual descriptions of the block (see the respective programmer instructions).
***

***** Block B_E2: business models *****

[Programmer information. Please ask the questions of this block once for every response in question E1. Please see the introduction for the order of the questions.]

[Title of the block:] Funding and business models

[if the respondent only ticked one answer in E1, the description of the block is:] In the next section, we would like to ask you about your business models.

[if the respondent ticked more than one answer in E1, the description of the block is:] In the next section, we would like to ask you about your business models. Please answer the questions with respect to the following service you provide: [*insert the first/second/third ticked response category of E1*].

E18. For which scientific discipline(s) does your organisation provide services?[13]

["Hilfetext":] Please follow this link for information on the classification of disciplines: https://en.wikipedia.org/wiki/Fields_of_Science_and_Technology]

[multiple choice]

- ☐ 1 - Natural Sciences
- ☐ 2 - Engineering and Technology
- ☐ 3 - Medical and Health Sciences
- ☐ 4 - Agricultural Sciences
- ☐ 5 - Social Sciences
- ☐ 6 - Humanities
- ☐ 99 - none/ not applicable [exclusive category, recoded in all other categories]
- ☐ 88 - other, please specify:

[13] Source of the categories (unchanged): Organisation for Economic Co-operation and Development (2007): Revised Field of Science and Technology Classification in the Frascati Manual; available at: https://www.oecd.org/science/inno/38235147.pdf; last accessed 6 November 2019.

E19. Who recurrently provides funding to your organisation?

[multiple choice]

- ◻ 1 - research institution(s)
- ◻ 2 - university
- ◻ 3 - state/ ministry
- ◻ 4 - region/ town
- ◻ 5 - research communities
- ◻ 6 - European funds
- ◻ 7 - industry/ small and medium-sized enterprises (SMEs)
- ◻ 8 - funding agencies/ funding bodies
- ◻ 88 - other, please specify:

E20. Does your organisation acquire own revenues other than funding?

[single choice]

- ◻ 2 - yes
- ◻ 1 - no
- ◻ 77 - don't know

E201 What are the sources of your own revenues other than funding?

[multiple choice, Condition: ask only if E20 == „2 - yes“]

- ◻ 1 - managed online services (software as a service, applications, storage, computing)
- ◻ 2 - hosting (hardware and services for third parties)
- ◻ 3 - consultancy or training
- ◻ 88 - other, please specify:

E21. Some organisations grant all users access to their services, some limit access to their services based on certain criteria. Does your organisation restrict access to its services to one or more of the following groups?

[multiple choice]

- ◻ 1 - users or communities approved by the funding body (e.g. due to regional or research topic restrictions)
- ◻ 2 - users selected by competition
- ◻ 3 - members of certain communities or organisations (e.g. virtual organisations)
- ◻ 4 - national users
- ◻ 5 - no access restrictions
- ◻ 99 - not applicable [exclusive category, recoded in all other categories]
- ◻ 88 - other, please specify:

E22. Does your organisation currently have policies, procedural and/ or technical barriers that limit the expansion of your services to further user groups?

[single choice]

- ◻ 2 - yes
- ◻ 1 - no
- ◻ 77 - don't know

E23. Please describe which policies, procedural and/ or technical barriers limit the expansion of your services to further user groups:

[long free text, Condition: ask only if E22 == „2 - yes"]

E25. Does your organisation charge users/ clients for services?

[single choice, Condition: ask only if E20 == "2 - yes"]

- ◻ 3 - yes, for all services
- ◻ 2 - yes, for some services
- ◻ 1 - no

E251 How does your organisation charge users/ clients for services?

[multiple choice, Condition: ask only if E25 is NOT "1 - no"]

- ◻ 1 - based on a price list for individual items/ services (e.g. use of datasets, a certain storage capacity or virtual machines)
- ◻ 2 - based on a flat rate for different sets of items/ services
- ◻ 3 - solutions tailored to the needs of user/ clients
- ◻ 88 - other, please specify:

E27. How do you buy supplies, resources or services?

[multiple choice]

- ◻ 1 - buy or rent without tender
- ◻ 2 - buy or rent with tender
- ◻ 3 - buy or rent on pre-negotiated procurement/tender
- ◻ 88 - other, please specify:
- ◻ 99 - not applicable [exclusive category, recoded in all other categories]

E30. Does your organisation know the unit cost of your services? If yes, what is the granularity?

[single choice]

- ◻ 1 - yes, cost per service
- ◻ 2 - yes, cost per set of services
- ◻ 3 - not currently available, but could be calculated
- ◻ 4 - no, but in preparation
- ◻ 5 - no, not possible/ not foreseen

CE2. Would you like to give a comment on any of the questions above?

[long free text]

***** Block B_E3 on SLAs *****

[Programmer information. Please ask the questions of this block once for every response in question E1. Please see the introduction for the order of the questions.]

[Title of the block:] Service Level Agreements (SLAs)

[if the respondent only ticked one answer in E1, the description of the block is:] In the next questions, we would like to ask you about Service Level Agreements (SLAs) you may provide. [if the respondent ticked more than one answer in E1, the description of the block is:] In the next questions, we would like to ask you about Service Level Agreements (SLAs) you may provide. Please answer the questions with respect to the following service you provide: [*insert the first/second/third ticked response category of E1*].

E28. Does your organisation offer Service Level Agreements (SLAs)?

[single choice]

- ◻ 1 - yes, for all services
- ◻ 2 - yes, for some services
- ◻ 3 - no, but foreseen in the near future
- ◻ 4 - no, not foreseen in the near future
- ◻ 99 - not applicable

E29. Are you participating in a transnational organisation or federation that offers Service Level Agreements (SLAs) or similar contracts that are also binding for your organisation?

[single choice]

- ◻ 2 - yes
- ◻ 1 - no
- ◻ 77 - don't know

E32. What types of Service Level Agreements (SLAs) do you offer or will you offer in the future?

[single choice, Condition: ask only if E28 == ("1 - yes, for all services" OR "2 - yes, for some services" OR "3 - no, but foreseen in the near future"]

- ◻ 1 - several predefined types
- ◻ 2 - custom made type
- ◻ 3 - one-fits-all
- ◻ 88 - other, please specify:
- ◻ 99 - not applicable

E31. Have you encountered issues or barriers to establish Service Level Agreements (SLAs) with a community?

[single choice, ask only if E28 == ("1 - yes, for all services" OR "2 - yes, for some services" OR "3 - no, but foreseen in the near future"]

- ◻ 2 - yes, please specify: [provide text box]
- ◻ 1 - no
- ◻ 77 - don't know
- ◻ 99 -not applicable

CE3. Would you like to give a comment on any of the questions above?

[long free text]

***** Block B_E4: contact with users *****

[Programmer information. Please ask the questions of this block once for every response in question E1. Please see the introduction for the order of the questions]

[Title of the block:] Users

[if the respondent only ticked one answer in E1, the description of the block is:] Next, we would like to ask questions about your users.

[if the respondent ticked more than one answer in E1, the description of the block is:] Next, we would like to ask questions about your users. Please answer the questions with respect to the following service you provide: [*insert the first/second/third ticked response category of E1*].

E78. We would like to know, which user groups use your organisation's services. How frequently do the following groups use your services?

[array question]

	1 - never	2 - not very frequently	3 - frequently	4 - very frequently	99 - not applicable	77 - don't know
A - (researchers based at) universities	☐	☐	☐	☐	☐	☐
B - (researchers of) non-university research institutions	☐	☐	☐	☐	☐	☐
C - (researchers of) private, commercial institutions	☐	☐	☐	☐	☐	☐
D - governmental institutions (e.g. census bureaus)	☐	☐	☐	☐	☐	☐
E - students	☐	☐	☐	☐	☐	☐
F - citizen scientists	☐	☐	☐	☐	☐	☐
G - professionals						
H - other:	☐	☐	☐	☐	☐	☐

E34. Some organisations collect their users' opinions on how to optimise their services and gather feedback from users, some do not. Does your organisation collect feedback from users?

[multiple choice; question was recoded: variable E34_1/2/3 captures whether respondents chose any affirmative answer. If respondents did so, the variables E34_1/2/3_4, E34_1/2/3_3 E34_1/2/3_2 contain the information which option(s) they chose.]

- ◻ 4 - yes, by tools implemented for this purpose (e. g. feedback forms)
- ◻ 3 - yes, by user meetings and workshops
- ◻ 2 - yes, by discussions
- ◻ 1 - no, user feedback is not collected [exclusive category]
- ◻ 88 - yes, by other means:

E35. Does your organisation offer training?

[single choice]

- ◻ 2 - yes
- ◻ 1 - no

E351 Which form(s) of training does your organisation offer?

[multiple choice, Condition: ask only if E35 == „2 - yes“]

- ◻ 1 - face-to-face training (e.g., workshops, lectures)
- ◻ 2 - massive open online courses (MOOCs) or web tutorials
- ◻ 3 - online documentation
- ◻ 88 - others, please specify:

E352 In which language(s) does your organisation provide training?

[multiple choice, Condition: ask only if E35 == „2 - yes“]

- ◻ 1 - country's/regional language
- ◻ 2 - English
- ◻ 3 - other or several, please specify:

E353 For whom does your organisation offer training?

[multiple choice, Condition: ask only if E35 == „2 - yes“]

- ◻ 1 - for everyone interested
- ◻ 2 - for specific communities or a specific community
- ◻ 3 - for everyone in a certain geographical area
- ◻ 4 - for everyone affiliated to (a) defined organisation(s)
- ◻ 88 - others, please specify:

E36. How does your organisation organise user support?

[multiple choice; question was recoded: variable E36_1/2/3 captures whether respondents chose any affirmative answer. If respondents did so, the variables E36_1/2/3_3, E36_1/2/3_2 and E36_1/2/3_88 contain the information which option(s) they chose.]

- ◻ 3 - We support users via a centrally organised system (e.g. ticketing system, helpdesk).
- ◻ 2 - We support users individually without a centrally organised user support system.
- ◻ 1 - There is no user support. [exclusive category]
- ◻ 88 - Other, please specify:
- ◻ 99 - not applicable [exclusive category, recoded to E36_1/2/3]

E79. Does your organisation offer a website that describes your service(s)?

[single choice]

- ◻ 2 - yes
- ◻ 1 - no

E791. In which language(s) is this website available?

[multiple choice question, Condition: ask only if E79 == "2 - yes"]

- ◻ 1 - Dutch
- ◻ 2 - English
- ◻ 3 - French
- ◻ 4 - German
- ◻ 5 - Italian
- ◻ 88 - other:

E80. Does your organisation provide the following services?

[array question]

	2 - yes	1 - no
A - support for data management plans	◻	◻
B - advice on data management	◻	◻
C - support concerning legal issues	◻	◻

E81. From which additional services would the users of your organisation benefit?

[long free text]

CE4. Would you like to give a comment on any of the questions above?

[long free text]

Block B_E5: access and legal aspects

[Programmer information. Please ask the questions of this block once for every response in question E1. Please see the introduction for the order of the questions.]

[Title of the block:] Access conditions and legal aspects

[if the respondent only ticked one answer in E1, the description of the block is:] The following questions refer to your organisation's access conditions and several legal aspects (e.g. data protection).

[if the respondent ticked more than one answer in E1, the description of the block is:] The following questions refer to your organisation's access conditions and several legal aspects (e.g. data protection). Please answer the questions with respect to the following service you provide: [*insert the first/second/third ticked response category of E1*].

E38. Does your organisation have a publicly available access policy for services or data?

[single choice]

- ◻ 2 - yes
- ◻ 1 - no

E39. Is a publicly available access policy planned?

[single choice, Condition: ask only if E38 == „1 - no“]

- ◻ 1 - no
- ◻ 2 - yes, in less than 1 year
- ◻ 3 - yes, in 1 to 2 years
- ◻ 4 - yes, in more than 2 years
- ◻ 99 - not applicable

E50. How does your organisation authorize access to data?

[multiple choice]

- ◻ 1 - Group membership: Members of groups can access all files open to the group.
- ◻ 2 - Mapping of group membership to the local file system.
- ◻ 3 - Access to datasets is authorized individually.
- ◻ 4 - Service local authorization: users choose with whom they share files, (e.g. NextCloud or GoogleDocs).
- ◻ 5 - No access control, data is openly accessible.
- ◻ 88 - other, please specify:
- ◻ 99 - not applicable [exclusive category, recoded in all other categories]

E46. Does your organisation offer services that process personal data in research data?

[single choice]

- ◻ 2 - yes
- ◻ 1 - no

E47 Does your organisation offer services that process special categories of personal data in research data?

Personal data is defined as data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union membership, genetic or biometric data for the purpose of uniquely identifying a natural person, data concerning health, a natural person's sex life or sexual orientation.[14]

[All the text above is part of the question (not the "Hilfetext"), single choice, Condition: ask only if E46 is NOT "1 - no"]

- ◻ 2 - yes
- ◻ 1 - no

[help text]: Special categories of personal data according to Art. 9 GDPR.

[14] Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation) (Text with EEA relevance), Article 9, § 1.

E90. Under what conditions does your organisation disseminate research data?

[multiple choice, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

- ◻ 1 - public domain (Copyright not applicable, Public Domain Mark, Creative Commons: CC 0) [variable name: E901]
- ◻ 2 - creative commons licences for open reuse (CC BY (Attribution), CC BY-SA (Attribution-Share Alike)) [variable name: E902]
- ◻ 3 - creative commons licenses for restricted reuse (CC BY ND (Attribution-NoDerivatives), CC BY NC (Attribution-NonCommercial), and combinations including ND or NC) [variable name: E903]
- ◻ 4 - we use tailored licences [variable name: E904]
- ◻ 88 - other, please specify: [variable name: E9088]

[help text:] If you have a list of your available licenses, please enter a link to the description or include them into the field "other, please specify".

E54. Do you need implementation support outside of your organisation to federate your service to EOSC?

[single choice, Condition: ask only if EOSCfamiliar is NOT "4 - not familiar at all"]

- ◻ 2 - yes
- ◻ 1 - no
- ◻ 77 - don't know

CE5. Would you like to give a comment on any of the questions above?

[long free text]

*** Block B_E6: Policies, APIs, PIDs***

[Programmer information. Please ask the questions of this block once for every response in question E1. Please see the introduction for the order of the questions]

[Title of the block:] Policies and technical specifications

[if the respondent only ticked one answer in E1, the description of the block is:] In this section, we would like to ask you about policies at your organisation as well as some technical features.

[if the respondent ticked more than one answer in E1, the description of the block is:] In this section, we would like to ask you about policies at your organisation as well as some technical features. Please answer the questions with respect to the following service you provide: [*insert the first/second/third ticked response category of E1*].

E56. Some organisations have developed regulations related to their work processes, whereas other organisations have not. Has your organisation developed informal or formal regulations or publicly available policies that address the following aspects?

[array question]

	1 - no regulation	2 - informal regulation	3 - formal/ written regulation	4 - publicly available policy	99 - not applic-able
A - research data management (RDM)	☐	☐	☐	☐	☐
B - open research data	☐	☐	☐	☐	☐
C - long-term availability of research data	☐	☐	☐	☐	☐
D - compliance of data to the FAIR principles	☐	☐	☐	☐	☐
E - publication of data in a repository	☐	☐	☐	☐	☐
F - publication of data in a certified repository	☐	☐	☐	☐	☐
G - other, please specify:	☐	☐	☐	☐	☐

E58. Are your organisation's services accessible by an application programming interface (API)?[15]

[single choice]

- ◻ 4 - This feature is fully implemented.
- ◻ 3 - This feature is in the implementation phase.
- ◻ 2 - We are working on or have a theoretical concept for this feature.
- ◻ 1 - We have not considered this feature yet.
- ◻ 99 - not applicable

E60. Does your organisation assign or provide persistent identifiers (e.g. DOI, Handle)?[16]

[single choice]

- ◻ 4 - This feature is fully implemented.
- ◻ 3 - This feature is in the implementation phase.
- ◻ 2 - We are working on or have a theoretical concept for this feature.
- ◻ 1 - We have not considered this feature yet.
- ◻ 99 - not applicable

E61. Which persistent identifiers does your organisation assign?

[multiple choice, Condition: ask only if E60 == ("4 - This feature is fully implemented." OR "3 - This feature is in the implementation phase." OR „2 - We are working on or have a theoretical concept for this feature.")]

- ◻ 1 - DOI
- ◻ 2 - URN
- ◻ 3 - Handle
- ◻ 88 - other, please specify:
- ◻ 99 - not applicable [exclusive category, recoded in all other categories]

CE6. Would you like to give a comment on any of the questions above?

[long free text]

[15] Source of categories (adapted) for the questions E58, E60, E64, E65 and E57: Core Trust Seal (2018): Core Trustworthy Data Repositories Extended Guidance. available at: https://www.coretrustseal.org/wp-content/uploads/2017/01/20180629-CTS-Extended-Guidance-v1.1.pdf, last accessed: 7 November 2019, p. 4.

[16] Source of this question (adapted): Core Trust Seal (2018): Core Trustworthy Data Repositories Extended Guidance. available at: https://www.coretrustseal.org/wp-content/uploads/2017/01/20180629-CTS-Extended-Guidance-v1.1.pdf, last accessed: 7 November 2019, p. 24.

***** Block B_E7: FAIRness of data *****

[Programmer Information: the questions of this block should only be asked to respondents who meet this condition: E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

[Title of the block:] FAIRness of data

[if the respondent only ticked one answer in E1, the description of the block is:] Next, we would like to ask you about the FAIRness of your data holdings.

[if the respondent ticked more than one answer in E1, the description of the block is:] Next, we would like to ask you about the FAIRness of your data holdings. Please answer the questions with respect to the following service you provide: [*insert the first/second/third ticked response category of E1*].

E63. How FAIR do you consider your data holdings?

[single choice, Condition: ask only if FAIRfamiliar is NOT "4 - not familiar at all" AND if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

- ◻ 4 - very much
- ◻ 3 - somewhat
- ◻ 2 - not very much
- ◻ 1 - not at all
- ◻ 99 - not applicable

E64. Does your organisation provide a search feature for research data?[17]

[single choice, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

- ◻ 4 - This feature is fully implemented.
- ◻ 3 - This feature is in the implementation phase.
- ◻ 2 - We are working on or have a theoretical concept for this feature.
- ◻ 1 - We have not considered this feature yet.
- ◻ 99 - not applicable

[17] Source of question (adapted): Core Trust Seal (2018): Core Trustworthy Data Repositories Extended Guidance. available at: https://www.coretrustseal.org/wp-content/uploads/2017/01/20180629-CTS-Extended-Guidance-v1.1.pdf, last accessed: 7 November 2019, p. 24.

E65. Does your organisation provide a search feature for metadata?[18]

[single choice, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

- ◻ 4 - This feature is fully implemented.
- ◻ 3 - This feature is in the implementation phase.
- ◻ 2 - We are working on or have a theoretical concept for this feature.
- ◻ 1 - We have not considered this feature yet.
- ◻ 99 - not applicable

E66. Does your organisation use standardized/ controlled vocabularies for metadata?

[single choice, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

- ◻ 2 - yes
- ◻ 1 - no
- ◻ 99 - not applicable

E661 Please list the most important standardized/ controlled vocabularies your organisation uses in your metadata.

[multiple short free text options, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)." AND E66 == "2 - yes"]

[18] Source of question (adapted): Core Trust Seal (2018): Core Trustworthy Data Repositories Extended Guidance. available at: https://www.coretrustseal.org/wp-content/uploads/2017/01/20180629-CTS-Extended-Guidance-v1.1.pdf, last accessed: 7 November 2019, p. 24.

E67. In which language(s) does your organisation provide metadata? Please indicate what percentage of your metadata are available in the following languages. Please estimate if you do not know the exact percentages.

[all text above is part of the question (not of the "Hilfetext"), array question, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data).", variable names: E67A, E67B etc.]

	0%	1-25%	26-50%	51-75%	76- 99%	100%
A - Dutch	☐	☐	☐	☐	☐	☐
B - English	☐	☐	☐	☐	☐	☐
C - French	☐	☐	☐	☐	☐	☐
D - German	☐	☐	☐	☐	☐	☐
E - Italian	☐	☐	☐	☐	☐	☐
F - other:	☐	☐	☐	☐	☐	☐

E57. Does your organisation provide a data catalogue in a machine-readable format?[19]

[single choice, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

- ☐ 4 - This feature is fully implemented.
- ☐ 3 - This feature is in the implementation phase.
- ☐ 2 - We are working on or have a theoretical concept for this feature.
- ☐ 1 - We have not considered this feature yet.
- ☐ 99 - not applicable

CE7. Would you like to give a comment on any of the questions above?

[long free text, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

[19] Source of question (adapted): Core Trust Seal (2018): Core Trustworthy Data Repositories Extended Guidance. available at: https://www.coretrustseal.org/wp-content/uploads/2017/01/20180629-CTS-Extended-Guidance-v1.1.pdf, last accessed: 7 November 2019, p. 24.

***** Block B_E8: FAIRNESS of data II *****

[Programmer Information: the questions of this block should only be asked to respondents who meet this condition: E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

[Title of the block:] FAIRness of data

[if the respondent only ticked one answer in E1, the description of the block is:] Next, we would like to ask you about the FAIRness of your data holdings.

[if the respondent ticked more than one answer in E1, the description of the block is:] Next, we would like to ask you about the FAIRness of your data holdings. Please answer the questions with respect to the following service you provide: [*insert the first/second/third ticked response category of E1*].

E69. Some researchers or organisations are reluctant to share and/ or publish data, others are not. How concerned are your customers/ depositors about the following aspects?

[array question, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

	1 - not concerned at all	2 - somewhat concerned	3 - concerned	4 - very concerned	77 - don't know	99 - not applicable
A - lack of control over the usage of data	☐	☐	☐	☐	☐	☐
B - effort of preparing the data for publication	☐	☐	☐	☐	☐	☐
C - doubts about the depositor's benefit of sharing data	☐	☐	☐	☐	☐	☐
D - competitive disadvantage when sharing	☐	☐	☐	☐	☐	☐
E - data protection	☐	☐	☐	☐	☐	☐
F - intellectual property (e.g. copyright)	☐	☐	☐	☐	☐	☐
G - other:	☐	☐	☐	☐	☐	☐

E70. Does your organisation use unique identifiers for researchers in the metadata?

[multiple choice, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."; question was recoded: variable E70 captures whether respondents chose any affirmative answer. If respondents did so, the variables E701-E7088 contain the information which option(s) they chose.]

- ◻ 1 - yes, ORCID [variable name: E701]
- ◻ 2 - yes, ResearcherID [variable name: E702]
- ◻ 3 - yes, other: [variable name: E7088]
- ◻ 4 – no [exclusive category]
- ◻ 99 - not applicable [exclusive category, recoded to E70]

E71. Does your organisation implement measures for ensuring documentation about the origin and the changes made in data (i.e. data provenance)? [20]

[single choice, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

- ◻ 2 - yes
- ◻ 1 - no
- ◻ 99 - not applicable

E72. Which measures for ensuring documentation about the origin and the changes made in data (i.e. data provenance) does your organisation implement?

[multiple choice, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)." AND E71 == "2 - yes"]

- ◻ 1 - version control [variable name: E721]
- ◻ 2 - file integrity checks (e.g. checksums, fixity checks) [variable name: E722]
- ◻ 88 - other: [variable name: E7288]
- ◻ 99 - not applicable [exclusive category, recoded in all other categories]

[20] Source of question E71 and E72 (adapted): Core Trust Seal (2018): Core Trustworthy Data Repositories Extended Guidance. available at: https://www.coretrustseal.org/wp-content/uploads/2017/01/20180629-CTS-Extended-Guidance-v1.1.pdf, last accessed: 7.11.2019, p. 17.

E73. Did you complete any of the following certifications or audits between 2015-2019?

[array question, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

	3 - yes	2 - no, but in preparation	1 - no
A - Core Trust Seal (CTS)	◻	◻	◻
B - Data Seal of Approval (DSA)	◻	◻	◻
C - World Data System (WDS)	◻	◻	◻
E - ISO 16363 certification	◻	◻	◻
F - Nestor Seal	◻	◻	◻
G - Digital Repository Audit Method Based on Risk Assessment (DRAMBORA)	◻	◻	◻
H - Trustworthy Repositories Audit & Certification (TRAC)	◻	◻	◻
I - other:	◻	◻	◻

E74. What level of curation do you perform?[21]

[multiple choice, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

- ◻ 1 - We distribute the content as deposited. [variable name: E471]
- ◻ 2 - We perform basic curation (e.g. brief checking, addition of basic metadata or documentation). [variable name: E472]
- ◻ 3 - We perform enhanced curation (e.g. conversion to new formats, enhancement of documentation). [variable name: E473]
- ◻ 4 - We perform data-level curation (all changes made above and additional editing of deposited data for accuracy). [variable name: E474]

CE8. Would you like to give a comment on any of the questions above?

[long free text, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

[21] Source of question (adapted): Core Trust Seal (2018): Core Trustworthy Data Repositories Extended Guidance. available at: https://www.coretrustseal.org/wp-content/uploads/2017/01/20180629-CTS-Extended-Guidance-v1.1.pdf, last accessed: 7 November 2019, p. 8.

***** Block B_E9: Repositories *****

[Programmer Information: the questions of this block should only be asked to respondents who meet this condition: E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

[Title of the block:] Characteristics of repositories

[Description of the block:] Please provide information on your repository below.

E93. What type of digital data is archived in your repository?[22]

[multiple choice, please randomize the response categories; Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

- ◻ 1 – Numeric [variable name: E931]
- ◻ 2 – Text [variable name: E932]
- ◻ 3 - Still image [variable name: E933]
- ◻ 4 – Geospatial [variable name: E934]
- ◻ 5 – Audio [variable name: E935]
- ◻ 6 – Video [variable name: E936]
- ◻ 7 – Software [variable name: E937]
- ◻ 8 - Interactive resource [variable name: E938]
- ◻ 9 - 3D [variable name: E939]
- ◻ 88 - Other: [variable name: E9388]

E76. How many datasets are accessible online? A dataset is a logical collection of one or more files which is described by a metadata scheme.

[short free text, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

E77. Please think about datasets that are reused frequently. What characteristics are important for a frequent reuse of datasets?

[long free text, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

[22] Source of categories (unchanged): Data Documentation Initiative (2016): DDI Controlled Vocabulary for Kind of Data Format, available at: https://ddialliance.org/Specification/DDI-CV/KindOfDataFormat_1.0.html, last accessed 8 November 2019.

CE9. Would you like to give a comment on any of the questions above?

[long free text, Condition: ask only if E1 == "1 - We offer data infrastructures which store and manage data (e.g. archive and disseminate data)."]

*** Block B_E11: Budget ***

[Programmer information: Please ask respondents the questions of Block B_E11 only ONCE, irrespective of the number of responses they gave in E1.]

[Title of the block:] Budget

[Description of the block:] The following questions aim at information on budget and staff of your organisation to provide context for the previous questions. They will allow comparisons between different types of organisations. Therefore, it is important to give estimates if you do not know the exact figures.

E2. What was your organisation's total budget (in €) in 2018?

[long free text]

E3. What was your organisation's budget (in €) in 2018 for the individual services you provide? Please list the budget for every service you provide.

[*** Condition: only ask this question if the respondents ticked more than one answer in question E1.

Responses: Please show the respondents the answers they ticked in question E1 and a corresponding text field where they can insert a number (ideally one text field per ticked response category of E1 (e.g. multiple choice with free text)).***]

E4. How many employees did your organisation have in 2018? Please provide a number in full time equivalents.

[long free text]

E5. How many employees did your organisation have in 2018 for the individual services you provide? Please provide a number in full time equivalents for every service. If you do not know the exact number, please estimate.

[*** Condition: only ask this question if the respondents ticked more than one answer in question E1.

Responses: Please show the respondents the answers they ticked in question E1 and a corresponding text field where they can insert a number (ideally one text field per ticked response category of E1 (e.g. multiple choice with free text)).***]

E6. What was the estimated gender distribution in your staff in 2018? Please provide the percentage of women.

[long free text]

E7. What was the estimated gender distribution in your staff in 2018 for the individual services you provide? Please provide the percentage of women for every service.

[*** Condition: only ask this question if the respondents ticked more than one answer in question E1.

Responses: Please show the respondents the answers they ticked in question E1 and a corresponding text field where they can insert a number (ideally one text field per ticked response category of E1 (e.g. multiple choice with free text))***]

E831. Which term(s) fit(s) your organisation best?

[single choice]

- ◻ 1- research infrastructure
- ◻ 2 - e-infrastructure
- ◻ 3 - both terms are equally valid
- ◻ 88 - neither term fits my organisation, the correct term is: [= “other”-option in Limesurvey”]

CE11. Would you like to give a comment on any of the questions above?

[long free text]

***** Block B_E12: technical questions *****

[Programmer information. Please ask the questions of this block once for every response in question E1. Please see the introduction for the order of the questions.]

[Title of the block:] Authentication and Authorization Infrastructure (AAI)

[if the respondent only ticked one answer in E1, the description of the block is:] The following questions concern the Authentication and Authorization Infrastructure (AAI) and are rather technical. We would kindly ask you to consult colleagues responsible for these technical questions if sensible.

[if the respondent ticked more than one answer in E1, the description of the block is:] The following questions concern the Authentication and Authorization Infrastructure (AAI) and are rather technical. We would kindly ask you to consult colleagues responsible for these technical questions if sensible. Please answer the questions with respect to the following service you provide: [*insert the first/second/third ticked response category of E1*].

E40. What is the authentication model of your service?

[multiple choice]

- ☐ 1 - Member of national federation
- ☐ 2 - Using EGI (Checkin)
- ☐ 3 - Using EUDAT (B2ACCESS)
- ☐ 4 - Local authentication (etc/passwd)
- ☐ 88 - other, please specify:

E41. Does your organisation plan to authenticate your service(s) through an Identity Provider (IdP)?

[single choice, Condition: ask only if E40 == “4 - Local authentication (e.g. passwd)”]

- ☐ 1 - no
- ☐ 2 - yes, in less than 1 year
- ☐ 3 - yes, in 1 to 2 years
- ☐ 4 - yes, in more than 2 years
- ☐ 99 - not applicable
- ☐ 77 - don’t know

E43. Is your service proxied to eduGAIN?

[single choice, Condition: ask only if E40 == ("1 - Member of national federation" OR "2 - Using EGI (Checkin)" OR "3 - Using EUDAT (B2ACCESS)")]

- ◻ 1 - Yes
- ◻ 2 - No, we don't use a proxy
- ◻ 3 - No, we use a different service provider-identity provider (SP-IdP) proxy: [= "other" option]
- ◻ 77 - don't know

E44. In order for your service(s) to become part of a federation, these services cannot rely on locally managed attributes only. Is the authorisation information for your service(s) managed locally at the service level or received from an external attribute authority?

[single choice, Condition: ask only if E40 == ("1 - Member of national federation" OR "2 - Using EGI (Checkin)" OR "3 - Using EUDAT (B2ACCESS)")]

- ◻ 1 - attributes are managed locally
- ◻ 2 - attributes are received from an external authority
- ◻ 3 - both, managed locally and received from an external authority
- ◻ 99 - not applicable
- ◻ 77 - don't know

The REFEDS Research and Scholarship Entity Category (R&S) has been designed as a simple and scalable way for Identity Providers to release minimal amounts of required personal data to service providers.

E51. Do you use the REFEDS R&S entity category?

[All the text above is part of the question (not the „Hilfetext"), single choice, Condition: ask only if E40 == ("1 - Member of national federation" OR "2 - Using EGI (Checkin)" OR "3 - Using EUDAT (B2ACCESS)")]

- ◻ 1 - yes
- ◻ 2 - yes, in less than 1 year
- ◻ 3 - yes, in 1 to 2 years
- ◻ 4 - yes, in more than 2 years
- ◻ 5 - no
- ◻ 99 - not applicable
- ◻ 77 - don't know

E52. Does your organisation make use of one of the following authentication technologies?

[multiple choice]

- ◻ 1 - OIDC
- ◻ 2 - SAML
- ◻ 3 - OAuth2
- ◻ 4 - X.509
- ◻ 88 - other, please specify:
- ◻ 99 - not applicable [exclusive category, recoded in all other categories]
- ◻ 77 - don't know [exclusive category, recoded in all other categories]

E92. Some organisations use assurance frameworks to ensure the quality and security of the digital identity of users, others use a different approach to reach this goal. Do you make use of any assurance framework?

[multiple choice; question was recoded: variable E92_1/2/3 captures whether respondents chose any affirmative answer. If respondents did so, the variables E92_1/2/3_1- E92_1/2/3_4 contain the information which option(s) they chose.]

- ◻ 1 - Yes, REFEDS Assurance framework
- ◻ 2 - Yes, Kantara IAF-SAC
- ◻ 3 - Yes, eIDAS
- ◻ 4 - Yes, NIST 800-63B "Authentication and Lifecycle management"
- ◻ 88 - Yes, we use a different assurance framework, please specify:
- ◻ 5 - No, we do not use assurance frameworks. [exclusive category, recoded to E92_1/2/3]

E48. Is there a security incident response procedure in place according to a certified framework?

[single choice]

- ◻ 2 - Yes, SIRTFI (Security Incident Response Trust Framework for Federated Identity)
- ◻ 88 - Yes, other, please specify:
- ◻ 1 - No
- ◻ 77 - don't know

E49. Is your resource provider, where your service(s) are hosted, or is your organisation ISO27001 certified?

[single choice]

- ◻ 1 - Yes, my resource provider is ISO27001 certified.
- ◻ 2 - Yes, my organisation is ISO27001 certified.
- ◻ 3 - No, neither is certified.
- ◻ 88 - No, but we use other certification or security standards, please specify: [=option "other"]
- ◻ 99 -not applicable
- ◻ 77 - don't know

E55. Does your organisation provide support for integration of your services in EOSC and for the accompanying implementation and upkeep of security standards and policies?

[multiple choice; question was recoded summarizing all response categories: 1 - "no", 2 - "yes, permanently", 3 - "yes, fixed term" , 4 - "yes, permanently and fixed term" , 77 - "don't know", 99 - "not applicable", 9999 - "no answer"]

- ◻ 1 - yes with dedicated staff (permanent)
- ◻ 2 - yes through local/national/international funded projects (fixed term)
- ◻ 3 - no
- ◻ 99 -not applicable
- ◻ 77 - don't know

CE12. Would you like to give a comment on any of the questions above?

[long free text]

*** Final question and closing message ***

[Show to all participants after their last questions]

C1. Would you like to receive the final report based on this survey by mail?

Please enter your e-mail address below:

[short text field]

Closing message

Thank you for participating in the EOSC pillar "National Initiatives" survey!

If you would like to subscribe to the newsletter, please visit our website: https://www.eosc-pillar.eu/

You can print your answers after submitting your responses.